Project Specification

Project Title

Empirical Market Sentiment Analysis & Asset Price Correlation Pipeline

1. Core System Architecture & Scope

The system is designed as an automated, offline computational pipeline that ingests financial headlines and daily stock market data for a target basket of equities, computes headline sentiment scores, calculates price returns, measures their statistical correlation, and generates visual analytics and summary reports.

2. Target Assets & Universe Selection

- **Stock Basket:** A configurable portfolio of 5–10 large-cap tickers (e.g., Apple AAPL, Microsoft MSFT, Reliance Industries RELIANCE.NS, TCS TCS.NS, Nvidia NVDA).
- **Time Horizon:**
  - **Intraday / Recent:** Past 7–30 days of daily trading sessions.
  - **Historical Price Window:** Aligned directly with the timestamp range of retrieved news articles.

3. Data Ingestion Requirements

3.1 Financial News Ingestion

- **Primary Source:** yfinance Ticker module news feed or public RSS financial feeds.
- **Extracted Attributes per Article:**
  - Article Title (String)
  - Publisher / Source (String)
  - Publication Timestamp (UTC, converted to YYYY-MM-DD HH:MM:SS)
  - Article URL (String)
  - Associated Ticker Symbol (String)

3.2 Market Price Ingestion

- **Primary Source:** Yahoo Finance API via yfinance.
- **Frequency:** Daily (1d interval).
- **Extracted Market Variables:**
  - Date (YYYY-MM-DD)
  - Open Price (P_{open})
  - High Price (P_{high})
  - Low Price (P_{low})
  - Close Price (P_{close})
  - Adjusted Close Price (P_{adj_close})
  - Volume (V)

4. Preprocessing & Data Cleaning Module

4.1 Text Cleaning & Standardization

- Strip HTML tags, escape entities, and remove non-printable characters.
- Preserve punctuation and capitalization patterns (critical for lexicon-based tools like VADER that utilize ALL-CAPS and exclamation marks for sentiment intensity).
- Deduplicate identical headlines published across multiple syndicates.

4.2 Time-Series Alignment

- **Timestamp Normalization:** Convert article publication timestamps to local exchange market time (e.g., EST for US markets, IST for NSE/BSE).
- **Non-Trading Day Shift Rule:**
  - If a news item is published after market close (e.g., after 4:00 PM EST) or on a weekend/holiday, map its publication date to the **next valid trading day ($t+1$)**.
- **Daily Aggregation:** Group multiple news headlines occurring on the same trading day t for ticker i.

5. Sentiment Quantification Engine

5.1 Lexicon Model

- **Engine:** NLTK VADER (SentimentIntensityAnalyzer).
- **Rationale:** Rule-based, computationally lightweight, handles financial slang/booster words, and requires zero model training.

5.2 Output Sentiment Metrics per Headline

- **Positive Polarity Score** ($S_{pos} \in [0, 1]$)
- **Negative Polarity Score** ($S_{neg} \in [0, 1]$)
- **Neutral Polarity Score** ($S_{neu} \in [0, 1]$)
- **Normalized Compound Score** ($S_{compound} \in [-1.0, +1.0]$):
  - $S_{compound} \ge 0.05 \implies$ **Positive Sentiment**
  - $S_{compound} \le -0.05 \implies$ **Negative Sentiment**
  - $-0.05 < S_{compound} < 0.05 \implies$ **Neutral Sentiment**

5.3 Daily Aggregated Sentiment Index

For a ticker on day t with N published news headlines:

- **Daily Mean Compound Score ($\bar{S}_t$):**
  $$\bar{S}_t = \frac{1}{N} \sum_{j=1}^{N} S_{compound, j}$$
- **Headline Volume (N_t):** Total count of news events on day t.

6. Financial Analytics & Feature Engineering

For each ticker on day t:

1. **Daily Log Return (R_t):**
   $$R_t = \ln\left(\frac{P_{close, t}}{P_{close, t-1}}\right)$$
2. **Daily Percentage Price Return (ΔP_t):**
   $$\Delta P_t = \frac{P_{close, t} - P_{close, t-1}}{P_{close, t-1}} \times 100$$
3. **Directional Target Variable (Y_t):**
   $$Y_t = \begin{cases} 1 & \text{if } \Delta P_t > 0 \text{ (Up)} \\ 0 & \text{if } \Delta P_t \le 0 \text{ (Down/Flat)} \end{cases}$$
4. **Intraday Volatility (σ_t):**
   $$\sigma_t = \frac{P_{high, t} - P_{low, t}}{P_{low, t}}$$

7. Statistical Analysis & Hypothesis Testing

1. **Pearson & Spearman Rank Correlation:**
   - Measure linear and monotonic correlation between Daily Mean Sentiment ($\bar{S}_t$) and Next-Day Return (R_{t+1}) or Same-Day Return (R_t).

2. **Directional Accuracy Test:**
   - Calculate what percentage of positive sentiment days ($\bar{S}_t > 0.05$) result in positive price returns ($R_t > 0$).
3. **Baseline Classification Metrics (scikit-learn):**
   - Train a simple **Logistic Regression** model using $\bar{S}_t$ and N_t to predict directional movement Y_t.
   - **Metrics to output:** Precision, Recall, F1-Score, ROC-AUC Score, and Confusion Matrix.

8. Data Visualization & Reporting Requirements

Generate and export static high-resolution charts (Matplotlib / Seaborn):

1. **Dual-Axis Time-Series Chart:**
   - **Primary Y-Axis (Left):** Daily Closing Stock Price (P_{close}).
   - **Secondary Y-Axis (Right):** Daily Compound Sentiment Score ($\bar{S}_t$) plotted as a bar/stem plot (Green for Positive, Red for Negative).
2. **Scatter Plot with Trend Line:**
   - **X-Axis:** Daily Mean Sentiment Score ($\bar{S}_t$).
   - **Y-Axis:** Daily Stock Price Return (ΔP_t).
   - **Overlay:** OLS Regression line showing the overall trend.
3. **Confusion Matrix Heatmap:**
   - Displaying True Positives, False Positives, True Negatives, and False Negatives for the directional prediction model.
4. **Sentiment Category Distribution:**
   - Donut chart showing the proportion of Positive vs. Negative vs. Neutral news headlines across the basket.

9. Data Output Formats & Persistence

- **Raw Data Store:** Saved as data/raw_news.csv and data/raw_prices.csv.
- **Processed Analytical Master Table:** Saved as data/master_sentiment_prices.csv containing columns:
  [Ticker, Date, Headline_Count, Mean_Sentiment, Sentiment_Class, Close_Price, Daily_Return_Pct, Price_Direction]
- **Exported Artifacts:** PNG/PDF charts saved in the outputs/figures/ directory.

10. Error Handling & Edge Cases

1. **Zero News Days:** Days with no news headlines must be imputed with a neutral sentiment score ($\bar{S}_t = 0.0$) and a flag no_news = True.
2. **API/Network Throttling:** Implement exponential backoff / retry mechanism for price/news fetching.
3. **Missing Market Data:** Handle trading holidays and weekend gaps cleanly without creating NaN values in return calculations.
4. **Text Noise:** Filter out non-financial headlines or promotional press releases.

11. Repository Structure Standards (GitHub Readiness)

```
financial-sentiment-price-tracker/
├── data/
│   ├── raw/
│   └── processed/
├── outputs/
│   ├── figures/
│   └── reports/
├── src/
│   ├── data_fetcher.py
│   ├── text_processor.py
│   ├── sentiment_engine.py
│   ├── financial_analytics.py
│   └── visualizer.py
├── notebooks/
│   └── exploratory_analysis.ipynb
├── tests/
│   └── test_sentiment.py
├── .gitignore
├── LICENSE
├── README.md
└── requirements.txt
```

12. README Specifications for Portfolio Impact

The README.md must contain:

1. **Executive Summary:** Brief overview of the empirical objective.
2. **System Architecture Diagram:** ASCII or graphic flowchart illustrating the pipeline.
3. **Key Findings Section:** Embedded chart images showing sentiment-return correlations.
4. **Reproducibility Guide:** Clear terminal commands to install dependencies, run the pipeline, and reproduce figures.
5. **Limitations & Future Scope:** Discussing lexicon limitations (e.g., VADER vs. FinBERT domain specificity) and lag effect modeling.